Conditional and Independent-Event Probabilities
Answer Key

Precalculus / Advanced Statistics — Discrete Mathematics

Quick Answers

1. $\frac{3}{5}$
2. $\frac{2}{5}$
3. $\frac{3}{5}$, $\frac{6}{25}$, 0.24
4. 0.9604
5. $\frac{5}{33}$
6. $\frac{2}{5}$
7. 0.2, 0.5
8. 0.012, 0.032
9. $a^3 + 3a^2b + 3ab^2 + b^3$
10. 0.384

Curriculum

Level: Precalculus / Advanced Statistics — Discrete Mathematics
HSA-APR.C.5 – *problem 9*
HSS-CP.A–HSS-CP.B – *problems 1, 2, 3, 4, 5, 6, 7, 8, 10*
Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

4. *If they got 1.96: added the two probabilities instead of multiplying independent events*
5. *If they got 0.17: treated the draws as independent, using 5 out of 12 twice instead of reducing the second draw*
10. *If they got 0.128: forgot the factor of 3 counting which shot is missed*

1. P(**sport** | **bus**). Conditioning on "rides the bus" restricts the sample space to that row, whose total is 80 students; of those, 48 play a sport.

$$P(\text{sport} \mid \text{bus}) = \frac{48}{80}$$

Dividing numerator and denominator by 16 gives $\boxed{\frac{3}{5}}$, that is 0.6.

2. P(**bus** | **sport**). Now the condition is "plays a sport", so the sample space is that column, total 120, and the favourable count is again the 48 students in the overlap cell.

$$P(\text{bus} \mid \text{sport}) = \frac{48}{120} = \boxed{\frac{2}{5}}$$

The two answers differ because the same intersection count is divided by a different conditioning total (80 versus 120): $P(A \mid B)$ and $P(B \mid A)$ are different questions, and confusing them is the classic conditional-probability reversal error.

3. Independence test from the table. (a) The column total for "plays a sport" is 120 out of 200 students:

$$P(\text{sport}) = \frac{120}{200} = \boxed{\frac{3}{5}} = 0.6$$

(b) The overlap cell holds 48 of the 200 students:

$$P(\text{sport} \cap \text{bus}) = \frac{48}{200} = \boxed{\frac{6}{25}} = 0.24$$

(c) $P(\text{bus}) = 80/200 = 0.4$, so the product is

$$P(\text{sport}) \cdot P(\text{bus}) = 0.6 \times 0.4 = \boxed{0.24}$$

The product equals $P(\text{sport} \cap \text{bus}) = 0.24$, so the two events **are independent**: knowing a student rides the bus does not change the chance the student plays a sport (both are 0.6, as problem 1 already showed).

4. Two chips, each passing with probability 0.98, tested independently. Independence is stated, so the probability that both pass is the product of the two individual probabilities — no conditional adjustment is needed:

$$P(\text{both pass}) = 0.98 \times 0.98 = 0.98^2$$

$$\boxed{0.9604}$$

Adding the probabilities (1.96) is impossible for a probability, which is a quick check that multiplication is the right operation.

5. Two red pens from 12, without replacement. Because the first pen is not replaced, the second factor is a *conditional* probability:

$$P(\text{both red}) = P(\text{1st red}) \cdot P(\text{2nd red} \mid \text{1st red}) = \frac{5}{12} \times \frac{4}{11} = \frac{20}{132}$$

The second factor $4/11$ is the probability the second pen is red *given* the first was red: one red pen and one pen overall have been removed. Reducing $20/132$ by 4 gives $\boxed{\frac{5}{33}}$. Treating the draws as independent would give $(5/12)^2 \approx 0.17$, which is too large.

6. P(**bus** | **no sport**). The condition "does not play a school sport" restricts the sample space to that column, whose total is 80 — not 200, because only students meeting the condition can be chosen. Of those 80, the bus riders number 32:

$$P(\text{bus} \mid \text{no sport}) = \frac{32}{80} = \boxed{\frac{2}{5}} = 0.4$$

This equals $P(\text{bus}) = 0.4$, another sign of the independence found in problem 3.

7. $P(A) = 0.4$**,** $P(B) = 0.5$**,** $P(A \cap B) = 0.25$**.** (a) The independence test compares the intersection with the product of the marginals:

$$P(A) \cdot P(B) = 0.4 \times 0.5 = \boxed{0.2}$$

(b) The conditional probability comes from the definition:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.25}{0.5} = \boxed{0.5}$$

Since $P(A) \cdot P(B) = 0.2$ but $P(A \cap B) = 0.25$, the events are **not independent**. The same conclusion in conditional language: $P(A \mid B) = 0.5$ while $P(A) = 0.4$, so knowing B occurred raises the chance of A.

8. Two production lines. (a) Line 1 supplies 60% of parts and 2% of those are defective, so the multiplication rule with a conditional second factor gives

$$P(\text{line } 1 \cap \text{defective}) = P(\text{line } 1) \cdot P(\text{defective} \mid \text{line } 1) = 0.6 \times 0.02 = \boxed{0.012}$$

(b) A defective part comes either from line 1 or from line 2, and those two cases do not overlap, so add the two branch products:

$$P(\text{defective}) = 0.6 \times 0.02 + 0.4 \times 0.05 = 0.012 + 0.020 = \boxed{0.032}$$

Sense check: the overall defect rate 3.2% lies between the two line rates 2% and 5%, closer to 2% because line 1 supplies more parts.

9. Expanding $(a + b)^3$ **for three independent throws.** Each throw contributes a factor of a (make) or b (miss), and because the throws are independent the probability of any particular sequence is the product of its three factors. Expanding collects the sequences by how many makes they contain:

$$\boxed{(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3}$$

The term $3a^2b$ gives exactly two makes: a^2b is the probability of one particular order (for example make, make, miss), and the coefficient 3 counts the three orders in which the single miss can occur — MMS, MSM, SMM. Coefficients larger than 1 are counting arrangements, not changing any probability.

10. Exactly two makes in three throws with $P(\textbf{make}) = 0.8$. (a) Use the $3a^2b$ term with $a = 0.8$ and $b = 1 - 0.8 = 0.2$:

$$3 \times 0.8^2 \times 0.2 = 3 \times 0.64 \times 0.2 = \boxed{0.384}$$

Dropping the coefficient 3 gives 0.128, the probability of one specific order only.
(b) **Open explanation — grade against this model.** Multiplying $0.8 \times 0.8 \times 0.2$ treats each throw's probability as unaffected by the others, which is exactly the independence assumption: $P(\text{make on throw 2} \mid \text{make on throw 1}) = P(\text{make on throw 2}) = 0.8$. If the throws were not independent — say a make raised her confidence — the second and third factors would have to be conditional probabilities such as $P(\text{make} \mid \text{previous make})$, which are not 0.8, and the general multiplication rule $P(A \cap B) = P(A) \cdot P(B \mid A)$ would have to be used with those values instead; the binomial model itself would no longer apply. Accept any answer that names the independence assumption and says the later factors would become conditional probabilities.